QF620 Stochastic Modelling in Finance — Session 5 Reference Cheatsheet

Stochastic Differential Equations: SDEs and martingales via Itô, the Bachelier (arithmetic) model and its call formula, the Black–Scholes (geometric) model and its call formula, the Vasicek mean-reverting model

Built from Session 5 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 5: Stochastic Differential Equations" (Tee Chyng Wen, SMU, 16 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the slides; check the course outline. From this session you need: applying Itô's formula to decide whether a process is a martingale (zero drift); solving the arithmetic, geometric and Vasicek SDEs; deriving the Bachelier and Black–Scholes call prices by splitting the integral at x^* and completing the square; and the mean, variance and distribution of the Vasicek short rate via Itô isometry.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$W_t,\ \tilde{W}_t$	(independent) Brownian motions	random	[S5]
$S_t,\ S_0,\ S_T$	stock price now / at time 0 / at maturity	price	
K	strike	price	
T	maturity (years)	time	
r	continuously compounded risk-free rate	rate	discount factor e^{-rT}
σ	volatility: price units (Bachelier), percentage (Black–Scholes)	number	△! same letter, different units [S10]
V_c	European call price	price	[S7]
$(x)^+$	$\max(x, 0)$	—	payoff
$\Phi,\ \phi$	standard normal CDF, PDF	function	$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$
x^*	exercise threshold for the standard normal variable	number	[S7, S12]
$d_1,\ d_2$	Black–Scholes arguments	number	[S14]
r_t	short interest rate	rate	Vasicek [S15]
$\kappa,\ \theta$	mean-reversion speed; long-run mean	number	[S15]
log	natural logarithm	function	slides write log for ln

Notation trap: on [S15–S16], r_t is a random short rate, while in the option formulas r is a constant discount rate. σ in Bachelier is measured in price units ($\approx \sigma_{\text{BS}} \times S_0$), so the two models' σ values are not comparable.

0. Topic map

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PREREQUISITES
S1  normal MGF, completing the square;  S3 Brownian motion, martingales
S4  Itô's formula dY = (μf' + ½σ²f'')dt + σf' dW;  Itô isometry
  |
  ▼
SESSION 5: STOCHASTIC DIFFERENTIAL EQUATIONS
├─ HISTORY & MARTINGALE PROCESS  [S2–S5]
|   ├─ Bru & Yor on Doeblin; BM as Gaussian, Markov, martingale, indep. increments [S2–S3]
|   ├─ Itô calculus took 25 years to be accepted (1944–1969, McKean)          [S4]
|   └─ Six processes: Itô SDE, martingale iff zero drift                       [S5]
|
├─ ARITHMETIC: BACHELIER (1900)  [S6–S8]
|   ├─ S_T = S₀ + σW_T ~ N(S₀, σ²T); negative prices possible                 [S6]
|   ├─ V_c = e^{−rT} E[(S_T − K)⁺]; exercise when x > x* = (K − S₀)/(σ√T)     [S7]
|   └─ V_c = e^{−rT}[(S₀−K)Φ(d) + σ√T φ(d)], d = (S₀−K)/(σ√T); ATM σ√(T/2π)   [S8]
|
├─ GEOMETRIC: BLACK–SCHOLES (1973)  [S9–S14]
|   ├─ S_T = S₀ exp[(r − ½σ²)T + σW_T]; log S_T normal                       [S9]
|   ├─ Sample paths: Bachelier vs Black–Scholes                              [S10]
|   ├─ dS = rS dt + σS dW solved via Itô on log S                            [S11]
|   ├─ x* = [log(K/S₀) − (r − ½σ²)T]/(σ√T); completing the square            [S12–S13]
|   └─ V_c = S₀Φ(d₁) − Ke^{−rT}Φ(d₂)                                         [S14]
|
└─ MEAN REVERSION: VASICEK (1977)  [S15–S16]
    ├─ dr = κ(θ − r)dt + σ dW; Itô on e^{κt}r_t; explicit solution            [S15]
    └─ r_t ~ N(r₀e^{−κt} + θ(1 − e^{−κt}), σ²(1 − e^{−2κt})/(2κ))            [S16]
  |
  ▼
LATER: S6 implied volatility, smile, displaced diffusion, stochastic volatility (Heston, SABR)
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Dependency logic:

- Itô's formula turns "is it a martingale?" into "is the dt coefficient zero?".

- An SDE is solved by finding a function that makes it linear: none needed for arithmetic BM, $\log S$ for geometric BM, $e^{\kappa t} r_t$ for Vasicek.
- Once S_T is a function of one normal variable, any European price is a one-dimensional integral; the call integral splits at the exercise boundary x^* into a Φ term and a ϕ (or second Φ) term.
- The Itô isometry from Session 4 gives the Vasicek variance in one line.

1 Concept entries

1.1 History of stochastic calculus [S2–S4]

- Bru and Yor (2002), commenting on Wolfgang Doeblin: the study of stochastic processes took a tortuous path through the 20th century; pioneers such as Bachelier, Lévy and Itô were original and took long to be accepted [S2].
- Brownian motion belongs to many classes at once: Gaussian processes, processes with independent increments, Markov processes, martingales and semimartingales [S3]. Specialists in each class made the field hard for outsiders.
- Itô's calculus took 25 years (1944–1969) to be accepted, the latter being the year of McKean's book *Stochastic Integrals* [S4].

1.2 SDEs and martingales [S5]

Rule *[Beyond slides]*: write $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ with Itô's formula. X is a martingale only if the drift $a_t = 0$ (then $X_t = X_0 + \int b\,\mathrm{d}W$, whose expected increment is 0 by Session 4).

$$\mathrm{d}g(t, W_t) = \Big(g_t + \tfrac{1}{2} g_{xx}\Big)\mathrm{d}t + g_x\,\mathrm{d}W_t.$$

where

$g(t,x)$	function of time and the Brownian value.
$g_t,\ g_x,\ g_{xx}$	partial derivatives in time, in x, and twice in x, evaluated at (t, W_t).
$\frac{1}{2} g_{xx}$	Itô term: W has $\mu = 0$, $\sigma = 1$ so $(\mathrm{d}W)^2 = \mathrm{d}t$.
$g_t + \frac{1}{2} g_{xx}$	the drift; it must vanish for a martingale.

For two independent Brownian motions, add one $\frac{1}{2} g_{yy}$ term and a $g_y\,\mathrm{d}\tilde{W}$ term ($\mathrm{d}W\,\mathrm{d}\tilde{W} = 0$). The six slide examples (working filled in, *[Beyond slides]*):

Process X_t	SDE $\mathrm{d}X_t$	Martingale?
(1) W_t^2	$\mathrm{d}t + 2W_t\,\mathrm{d}W_t$	no (drift 1)
(2) $2 + t + e^{W_t}$	$\big(1 + \frac{1}{2} e^{W_t}\big)\mathrm{d}t + e^{W_t}\,\mathrm{d}W_t$	no
(3) $W_t^2 + \tilde{W}_t^2$	$2\,\mathrm{d}t + 2W_t\,\mathrm{d}W_t + 2\tilde{W}_t\,\mathrm{d}\tilde{W}_t$	no (drift 2)
(4) $W_t^2 - t$	$2W_t\,\mathrm{d}W_t$	yes
(5) W_t^3	$3W_t\,\mathrm{d}t + 3W_t^2\,\mathrm{d}W_t$	no (drift $3W_t$)
(6) $e^{\theta W_t - \frac{1}{2}\theta^2 t}$	$\theta X_t\,\mathrm{d}W_t$	yes

Worked (6): $g = e^{\theta x - \frac{1}{2}\theta^2 t}$, so

$$g_t = -\tfrac{1}{2}\theta^2 g, \quad g_x = \theta g, \quad g_{xx} = \theta^2 g \quad \Rightarrow \quad \mathrm{d}X_t = \Big(-\tfrac{1}{2}\theta^2 + \tfrac{1}{2}\theta^2\Big) X_t\,\mathrm{d}t + \theta X_t\,\mathrm{d}W_t = \theta X_t\,\mathrm{d}W_t.$$

where

θ	real constant.
$-\frac{1}{2}\theta^2 X_t$	from the explicit $-\frac{1}{2}\theta^2 t$ in the exponent.
$+\frac{1}{2}\theta^2 X_t$	Itô term from g_{xx}; the two cancel, so there is no drift.
$\theta X_t\,\mathrm{d}W_t$	pure diffusion, hence a martingale (the exponential martingale of Session 3).

△! *[Beyond slides]* Zero drift strictly gives a *local* martingale; it is a true martingale when $\mathbb{E}\int_0^T b_t^2\,\mathrm{d}t < \infty$, which holds for all slide examples.

1.3 Bachelier model: arithmetic Brownian motion [S6–S7]

Bachelier (1900) first used a random walk for stock prices. The price is a symmetric random walk, i.e. a market in **equilibrium** [S6]:

$$\mathrm{d}S_t = \sigma\,\mathrm{d}W_t \quad\Longrightarrow\quad S_T = S_0 + \sigma W_T, \qquad W_T \sim N(0,T), \qquad S_T \sim N(S_0,\ \sigma^2 T).$$

where

σ	absolute (normal) volatility, in price units per $\sqrt{\text{year}}$.
$\sigma\,\mathrm{d}W_t$	the only source of change: no drift.
S_T	normally distributed around today's price.
$\sigma^2 T$	variance grows linearly in time.

Shortcoming [S6]: no lower bound at 0, so negative stock prices have positive probability. Reasonable for interest rates (which can be negative).

Call price set-up [S7]: write $W_T = \sqrt{T}\,x$ with $x \sim N(0,1)$:

$$V_c = e^{-rT}\mathbb{E}\big[(S_T-K)^+\big] = \frac{e^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\big(S_0+\sigma\sqrt{T}x-K\big)^+ e^{-\frac{x^2}{2}}\,\mathrm{d}x.$$

where

e^{-rT}	discount factor from T back to today.
$(S_T-K)^+$	call payoff.
x	standard normal variable; $\frac{1}{\sqrt{2\pi}}e^{-x^2/2} = \phi(x)$ is its density.
$S_0+\sigma\sqrt{T}x$	S_T written in terms of x.

The payoff is non-zero only when

$$S_0+\sigma\sqrt{T}x-K>0 \quad\Longrightarrow\quad x > \frac{K-S_0}{\sigma\sqrt{T}} = x^*.$$

where

x^*	the critical value of x above which the call finishes in the money.
$K-S_0$	how far the strike is above today's price.
$\sigma\sqrt{T}$	standard deviation of S_T; dividing standardises the distance.

△**!** [S6–S7] The model has zero drift but discounts at r. That is only risk-neutral if $r=0$, or if S is read as the *forward* price (then e^{-rT} is the bond price). Under $\mathbb{Q}$ with a spot stock, the drift should be rS_t *[Beyond slides]*.

1.4 Bachelier call formula [S8]

Split the integral above x^* into a constant part and a linear part:

$$V_c = \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^{\infty}(S_0-K)e^{-\frac{x^2}{2}}\,\mathrm{d}x + \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^{\infty}\sigma\sqrt{T}\,x\,e^{-\frac{x^2}{2}}\,\mathrm{d}x = e^{-rT}\Big[(S_0-K)\Phi(-x^*) + \frac{\sigma\sqrt{T}}{\sqrt{2\pi}}e^{-\frac{(x^*)^2}{2}}\Big].$$

where

$\frac{1}{\sqrt{2\pi}}\int_{x^*}^{\infty} e^{-x^2/2}\mathrm{d}x$	$= 1-\Phi(x^*) = \Phi(-x^*)$: the probability of finishing in the money.
$\int_{x^*}^{\infty} xe^{-x^2/2}\mathrm{d}x$	$= \big[-e^{-x^2/2}\big]_{x^*}^{\infty} = e^{-(x^*)^2/2}$, since $\frac{\mathrm{d}}{\mathrm{d}x}(-e^{-x^2/2}) = xe^{-x^2/2}$.

Since $-x^* = \frac{S_0-K}{\sigma\sqrt{T}}$ and ϕ is symmetric:

$$V_c = e^{-rT}\Big[(S_0-K)\,\Phi\Big(\frac{S_0-K}{\sigma\sqrt{T}}\Big) + \sigma\sqrt{T}\,\phi\Big(\frac{S_0-K}{\sigma\sqrt{T}}\Big)\Big], \qquad V_c^{\text{ATM}} = e^{-rT}\sigma\sqrt{\frac{T}{2\pi}}\ \ (K=S_0).$$

where

$\Phi(\cdot)$	standard normal CDF.
$\phi(\cdot)$	standard normal PDF, $\phi(0) = 1/\sqrt{2\pi} \approx 0.3989$.
$\frac{S_0-K}{\sigma\sqrt{T}}$	moneyness in standard deviations ($= -x^*$).
$(S_0-K)\Phi(\cdot)$	intrinsic value weighted by the probability of exercise.
$\sigma\sqrt{T}\,\phi(\cdot)$	time value from the spread of outcomes; the whole price at the money.

[Beyond slides] Put by parity: $V_p = V_c - e^{-rT}(S_0-K)$, i.e. $e^{-rT}\big[(K-S_0)\Phi(-d)+\sigma\sqrt{T}\phi(d)\big]$, $d = \frac{S_0-K}{\sigma\sqrt{T}}$. ATM rule of thumb: $V \approx 0.4\,\sigma\sqrt{T}$.

1.5 Black–Scholes model: geometric Brownian motion [S9–S10]

Black and Scholes (1973) [S9]:

$$S_T = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T\right], \qquad W_T \sim N(0,T), \qquad \mathbb{E}[S_T] = S_0 e^{rT}.$$

where

r	risk-free rate: under $\mathbb{Q}$ the stock grows at r on average.
σ	percentage (lognormal) volatility.
$-\frac{\sigma^2}{2}T$	convexity correction so that $\mathbb{E}[e^{\sigma W_T}] = e^{\frac{1}{2}\sigma^2 T}$ is cancelled.
$\mathbb{E}[S_T] = S_0 e^{rT}$	check with the normal MGF (Session 1).

Equivalently [S9]:

$$\log\frac{S_T}{S_0} \sim N\left(\left(r - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right), \qquad \log S_T = \log S_0 + \left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,N(0,1).$$

where

$\log\frac{S_T}{S_0}$	continuously compounded return over $[0,T]$: normal.
$(r - \frac{1}{2}\sigma^2)T$	mean log return.
$\sigma^2 T$	variance of the log return.

Sample paths compared [S10] (same Brownian path):

$$\text{Black–Scholes: } S_{t+\Delta t} = S_t \exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma(W_{t+\Delta t} - W_t)\right], \qquad \text{Bachelier: } S_{t+\Delta t} = S_t + \sigma(W_{t+\Delta t} - W_t).$$

where

Δt	simulation step.
$W_{t+\Delta t} - W_t$	shared Brownian increment, $\sim N(0, \Delta t)$.
multiplicative vs additive	Black–Scholes moves are proportional to the price level; Bachelier moves are the same size at any level.

Slide question [S10] (answer *[Beyond slides]*): with matched volatility at the start, the paths track closely while S is near S_0. When S rises, Black–Scholes moves get bigger; when S falls, they shrink and S stays positive, whereas Bachelier keeps the same absolute size and can go negative. Black–Scholes returns are lognormal and right-skewed; Bachelier outcomes are symmetric.

△**!** [S10] uses the same σ in both recursions. To be comparable, the Bachelier σ must be in price units, roughly $\sigma_{\text{BS}} S_0$.

1.6 Solving the Black–Scholes SDE [S11]

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t.$$

where

$rS_t\,\mathrm{d}t$	drift: return at the risk-free rate (risk-neutral measure).
$\sigma S_t\,\mathrm{d}W_t$	noise proportional to the price: constant percentage volatility.

Direct integration $S_T = S_0 + \int_0^T rS_t\,\mathrm{d}t + \int_0^T \sigma S_t\,\mathrm{d}W_t$ does not solve it (S appears on both sides). Apply Itô to $X_t = f(S_t) = \log S_t$ ($f' = \frac{1}{S}$, $f'' = -\frac{1}{S^2}$):

$$\mathrm{d}X_t = \left(rS_t \cdot \frac{1}{S_t} + \frac{1}{2}\sigma^2 S_t^2 \cdot \left(-\frac{1}{S_t^2}\right)\right)\mathrm{d}t + \sigma S_t \cdot \frac{1}{S_t}\,\mathrm{d}W_t = \left(r - \frac{\sigma^2}{2}\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

where

$\mu_t f'$	$rS \cdot \frac{1}{S} = r$.
$\frac{1}{2}\sigma_t^2 f''$	$\frac{1}{2}\sigma^2 S^2 \cdot (-\frac{1}{S^2}) = -\frac{1}{2}\sigma^2$: the Itô correction.
$\sigma_t f'$	$\sigma S \cdot \frac{1}{S} = \sigma$.
$\mathrm{d}X_t$	constant coefficients: $\log S$ is an arithmetic Brownian motion.

Integrate from 0 to T and exponentiate:

$$X_T - X_0 = \Big(r - \frac{\sigma^2}{2}\Big)T + \sigma W_T \quad \Longrightarrow \quad S_T = S_0 \exp\Big[\Big(r - \frac{\sigma^2}{2}\Big)T + \sigma W_T\Big].$$

where

$X_0 = \log S_0$	initial log price.
$W_0 = 0$	so $\int_0^T \mathrm{d}W = W_T$.

1.7 Black–Scholes call: setting up [S12]

With $W_T = \sqrt{T}x$, $x \sim N(0,1)$:

$$V_c = e^{-rT}\mathbb{E}[(S_T - K)^+] = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Big(S_0 e^{(r-\frac{\sigma^2}{2})T + \sigma\sqrt{T}x} - K\Big)^+ e^{-\frac{x^2}{2}}\,\mathrm{d}x.$$

where

$S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}x}$	S_T as a function of the standard normal x.
$e^{-x^2/2}/\sqrt{2\pi}$	density of x.

Exercise region:

$$S_T - K > 0 \quad \Longrightarrow \quad x > \frac{\log\left(\frac{K}{S_0}\right) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = x^*.$$

where

$\log(K/S_0)$	log-distance from the spot up to the strike.
$(r - \frac{1}{2}\sigma^2)T$	expected log drift, which works in favour of exercise.
$\sigma\sqrt{T}$	standard deviation of the log return.
x^*	so $\mathbb{P}^*(\text{exercise}) = \Phi(-x^*) = \Phi(d_2)$.

So

$$V_c = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} \Big(S_0 e^{(r-\frac{\sigma^2}{2})T + \sigma\sqrt{T}x} - K\Big) e^{-\frac{x^2}{2}}\,\mathrm{d}x.$$

where

$\int_{x^*}^{\infty}$	on this region the $(\cdot)^+$ can be dropped.

1.8 Black–Scholes call: completing the square [S13–S14]

Stock term: combine exponents and complete the square [S13]:

$$\frac{S_0 e^{-\frac{\sigma^2 T}{2}}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{x^2 - 2\sigma\sqrt{T}x}{2}}\,\mathrm{d}x = \frac{S_0 e^{-\frac{\sigma^2 T}{2}}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x-\sigma\sqrt{T})^2}{2}} e^{\frac{\sigma^2 T}{2}}\,\mathrm{d}x = \frac{S_0}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x-\sigma\sqrt{T})^2}{2}}\,\mathrm{d}x.$$

where

$e^{-rT} \cdot e^{rT}$	cancel, leaving $S_0 e^{-\frac{1}{2}\sigma^2 T}$ in front.
$x^2 - 2\sigma\sqrt{T}x$	$= (x - \sigma\sqrt{T})^2 - \sigma^2 T$: add and subtract $\sigma^2 T$.
$e^{\frac{1}{2}\sigma^2 T}$	the leftover constant; it cancels the $e^{-\frac{1}{2}\sigma^2 T}$ in front.
$(x - \sigma\sqrt{T})^2$	a normal density shifted to mean $\sigma\sqrt{T}$.

Evaluate both integrals [S14]:

$$V_c = S_0\big[\Phi(\infty) - \Phi(x^* - \sigma\sqrt{T})\big] - Ke^{-rT}\Phi(-x^*) = S_0\Phi(-x^* + \sigma\sqrt{T}) - Ke^{-rT}\Phi(-x^*).$$

where

$\Phi(-x^* + \sigma\sqrt{T})$	probability that a normal with mean $\sigma\sqrt{T}$ exceeds x^*: substitute $y = x - \sigma\sqrt{T}$.
$Ke^{-rT}\Phi(-x^*)$	discounted strike times the risk-neutral exercise probability.

$$V_c = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_1 = \frac{\log\frac{S_0}{K} + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} = \frac{\log\frac{S_0}{K} + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.$$

where

S_0	spot price.
K	strike.
$d_2 = -x^*$	standardised distance to exercise; $\Phi(d_2) = \mathbb{Q}(S_T > K)$.
$d_1 = d_2 + \sigma\sqrt{T}$	$\Phi(d_1)$ is the exercise probability under the stock measure, and the call delta $\partial V_c/\partial S_0$ *[Beyond slides].*
$S_0\Phi(d_1)$	present value of receiving the stock when exercised.
$Ke^{-rT}\Phi(d_2)$	present value of paying the strike when exercised.

[Beyond slides] Put: $V_p = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$; parity $V_c - V_p = S_0 - Ke^{-rT}$. Black (forward) form: $V_c = e^{-rT}[F\Phi(d_1) - K\Phi(d_2)]$ with $F = S_0e^{rT}$, $d_1 = \frac{\log(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$.

1.9 Vasicek mean-reverting model [S15]

The Ornstein–Uhlenbeck process models gas molecules; Vasicek (1977) used it for interest rates:

$$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

where

r_t	short interest rate at time t.
θ	long-run mean level.
$\kappa > 0$	speed of mean reversion: the drift pulls r_t toward θ, harder the further away it is.
$\theta - r_t$	positive when the rate is below its mean (drift up), negative above (drift down).
σ	absolute volatility of the rate (normal model).

Solve: Itô on $X_t = e^{\kappa t}r_t = f(t, r_t)$ ($f_t = \kappa e^{\kappa t}r$, $f_r = e^{\kappa t}$, $f_{rr} = 0$):

$$\mathrm{d}(e^{\kappa t}r_t) = \kappa e^{\kappa t}r_t\,\mathrm{d}t + e^{\kappa t}\mathrm{d}r_t = \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t.$$

where

$e^{\kappa t}$	integrating factor: chosen so that the $-\kappa r_t$ terms cancel.
$\kappa e^{\kappa t}r_t\,\mathrm{d}t$	from f_t.
$e^{\kappa t}\mathrm{d}r_t$	from f_r; no Itô term because f is linear in r.
$\kappa\theta e^{\kappa t}\,\mathrm{d}t$	what remains of the drift: no longer depends on r.

Integrate from 0 to t and divide by $e^{\kappa t}$:

$$r_t = r_0e^{-\kappa t} + \theta\left(1 - e^{-\kappa t}\right) + \sigma\int_0^t e^{\kappa(u-t)}\,\mathrm{d}W_u.$$

where

$r_0e^{-\kappa t}$	memory of the starting rate, decaying exponentially.
$\theta(1 - e^{-\kappa t})$	pull toward the long-run mean; $\int_0^t \kappa\theta e^{\kappa u}\mathrm{d}u = \theta(e^{\kappa t} - 1)$.
$\sigma\int_0^t e^{\kappa(u-t)}\mathrm{d}W_u$	accumulated shocks, older ones damped by $e^{-\kappa(t-u)}$.
u	integration (time) variable.

1.10 Vasicek distribution [S16]

Mean (stochastic integral has mean 0):

$$\mathbb{E}[r_t] = r_0e^{-\kappa t} + \theta\left(1 - e^{-\kappa t}\right).$$

where

$\mathbb{E}[r_t]$	weighted average of the start r_0 and long-run θ; weight on θ grows from 0 to 1.

Variance via Itô isometry $\mathbb{E}\big[(\int_0^T X_t\,\mathrm{d}W_t)^2\big] = \mathbb{E}\big[\int_0^T X_t^2\,\mathrm{d}t\big]$:

$$V[r_t] = \mathbb{E}\Big[\Big(\sigma\int_0^t e^{\kappa(u-t)}\mathrm{d}W_u\Big)^2\Big] = \sigma^2\int_0^t e^{2\kappa(u-t)}\,\mathrm{d}u = \frac{\sigma^2}{2\kappa}\big(1-e^{-2\kappa t}\big).$$

where

$e^{\kappa(u-t)}$	deterministic integrand, so the isometry needs no expectation inside.
$\int_0^t e^{2\kappa(u-t)}\mathrm{d}u$	$= \frac{1}{2\kappa}(1-e^{-2\kappa t})$.

Distribution (deterministic integrand ⇒ normal):

$$r_t \sim N\Big(r_0e^{-\kappa t} + \theta\big(1-e^{-\kappa t}\big),\ \frac{\sigma^2}{2\kappa}\big(1-e^{-2\kappa t}\big)\Big).$$

where

$t\to\infty$	stationary distribution $N\big(\theta,\ \frac{\sigma^2}{2\kappa}\big)$ *[Beyond slides]*.
$\kappa\to 0$	variance $\to \sigma^2 t$: back to arithmetic Brownian motion *[Beyond slides]*.

[Beyond slides] Weakness: normal, so negative rates have positive probability (CIR $\mathrm{d}r = \kappa(\theta-r)\mathrm{d}t + \sigma\sqrt{r}\,\mathrm{d}W$ fixes this). Half-life of a deviation: $\ln 2/\kappa$.

1.11 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
def bachelier_call(S0, K, sigma, T, r=0.0):
    """Arithmetic BM dS = sigma dW; sigma in price units."""
    d = (S0 - K) / (sigma*np.sqrt(T))
    return np.exp(-r*T) * ((S0 - K)*norm.cdf(d) + sigma*np.sqrt(T)*norm.pdf(d))
def bs_call(S0, K, sigma, T, r):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return S0*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d2)
rng = np.random.default_rng(2); Z = rng.standard_normal(2_000_000)
S0, K, T, r = 100, 105, 1.0, 0.05
ST = S0 + 20*np.sqrt(T)*Z                                  # Bachelier, sigma_N = 20, r = 0
print(round(bachelier_call(S0, K, 20, T), 4), round(np.maximum(ST-K, 0).mean(), 4))  # 5.7269 5.7292
print(round(bachelier_call(100, 100, 20, 1), 4), round(20*np.sqrt(1/(2*np.pi)), 4))  # ATM 7.9788 7.9788
ST = S0*np.exp((r - 0.02)*T + 0.2*np.sqrt(T)*Z)            # Black-Scholes, sigma = 20%
print(round(bs_call(S0, K, 0.2, T, r), 4), round(np.exp(-r*T)*np.maximum(ST-K, 0).mean(), 4))  # 8.0214 8.0257
kappa, theta, sig, r0, t = 1.5, 0.04, 0.01, 0.02, 2.0      # Vasicek
mean = r0*np.exp(-kappa*t) + theta*(1 - np.exp(-kappa*t))  # 0.039
var = sig**2/(2*kappa)*(1 - np.exp(-2*kappa*t))            # 3.33e-5; Euler simulation agrees
```

2 Comparison tables

2.1 The three SDEs

	Bachelier [S6–S8]	Black–Scholes [S9–S14]	Vasicek [S15–S16]
SDE	$\mathrm{d}S = \sigma\,\mathrm{d}W$	$\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$	$\mathrm{d}r = \kappa(\theta-r)\mathrm{d}t + \sigma\,\mathrm{d}W$
Transform	none	$\log S$	$e^{\kappa t}r_t$
Solution	$S_0 + \sigma W_T$	$S_0e^{(r-\frac{1}{2}\sigma^2)T+\sigma W_T}$	$r_0e^{-\kappa t} + \theta(1-e^{-\kappa t}) + \sigma\int e^{\kappa(u-t)}\mathrm{d}W$
Distribution	normal	lognormal	normal
Mean	S_0	S_0e^{rT}	$r_0e^{-\kappa t} + \theta(1-e^{-\kappa t})$
Variance	σ^2T	$S_0^2e^{2rT}(e^{\sigma^2T}-1)$ *[Beyond slides]*	$\frac{\sigma^2}{2\kappa}(1-e^{-2\kappa t})$
Negative values	possible	impossible	possible
Used for	rates, spreads	equities, FX	short rates

2.2 Bachelier vs Black–Scholes call

	Bachelier	Black–Scholes
x^*	$\frac{K-S_0}{\sigma\sqrt{T}}$	$\frac{\log(K/S_0)-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$
Price	$e^{-rT}[(S_0-K)\Phi(d)+\sigma\sqrt{T}\phi(d)]$	$S_0\Phi(d_1)-Ke^{-rT}\Phi(d_2)$
Key step	$\int xe^{-x^2/2}=e^{-x^2/2}$	completing the square
ATM	$e^{-rT}\sigma\sqrt{T/2\pi}$	$\approx 0.4\,S_0\sigma\sqrt{T}$ when $r=0$ *[Beyond slides]*
σ units	price	percentage

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Itô and martingales [S5]

- $\mathrm{d}g(t,W)=(g_t+\frac{1}{2}g_{xx})\mathrm{d}t+g_x\,\mathrm{d}W$; martingale $\iff$ drift $=0$ (with integrability).
  where g_t, g_x, g_{xx} = partial derivatives.

Bachelier [S6–S8]

- $S_T=S_0+\sigma W_T\sim N(S_0,\sigma^2T)$; $x^*=\frac{K-S_0}{\sigma\sqrt{T}}$.
  where σ = normal vol in price units.
- $V_c=e^{-rT}\big[(S_0-K)\Phi(d)+\sigma\sqrt{T}\phi(d)\big]$, $d=\frac{S_0-K}{\sigma\sqrt{T}}$; ATM $e^{-rT}\sigma\sqrt{T/2\pi}$.
  where Φ,ϕ = normal CDF, PDF.

Black–Scholes [S9–S14]

- $\mathrm{d}S=rS\mathrm{d}t+\sigma S\mathrm{d}W$; $\mathrm{d}\log S=(r-\frac{1}{2}\sigma^2)\mathrm{d}t+\sigma\mathrm{d}W$; $S_T=S_0e^{(r-\frac{1}{2}\sigma^2)T+\sigma W_T}$.
  where r = rate; σ = percentage vol.
- $V_c=S_0\Phi(d_1)-Ke^{-rT}\Phi(d_2)$; $d_{1,2}=\frac{\log(S_0/K)+(r\pm\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$; $d_2=-x^*$.
  where $\Phi(d_2)$ = risk-neutral exercise probability; $\Phi(d_1)$ = delta.

Vasicek [S15–S16]

- $\mathrm{d}r=\kappa(\theta-r)\mathrm{d}t+\sigma\mathrm{d}W$; $r_t=r_0e^{-\kappa t}+\theta(1-e^{-\kappa t})+\sigma\int_0^t e^{\kappa(u-t)}\mathrm{d}W_u$.
  where κ = reversion speed; θ = long-run mean.
- $r_t\sim N\big(r_0e^{-\kappa t}+\theta(1-e^{-\kappa t}),\ \frac{\sigma^2}{2\kappa}(1-e^{-2\kappa t})\big)$.
  where variance from Itô isometry.

4 Glossary

- **Arithmetic Brownian motion:** $\mathrm{d}S=\mu\,\mathrm{d}t+\sigma\,\mathrm{d}W$; normal outcomes [S7].
- **Bachelier model:** arithmetic BM for prices (1900) [S6].
- **Black–Scholes model:** geometric BM for prices (1973) [S9].
- **Completing the square:** rewriting $x^2-2ax=(x-a)^2-a^2$ to integrate exponentials of quadratics [S13].
- **Exercise boundary x^*:** level of the standard normal above which the option pays [S7, S12].
- **Geometric Brownian motion:** $\mathrm{d}S=\mu S\,\mathrm{d}t+\sigma S\,\mathrm{d}W$; lognormal outcomes [S11].
- **Integrating factor:** multiplier ($e^{\kappa t}$) that removes the state from the drift [S15].
- **Mean reversion:** drift pulling a process toward a long-run level [S15].
- **Ornstein–Uhlenbeck process:** $\mathrm{d}X=\kappa(\theta-X)\mathrm{d}t+\sigma\mathrm{d}W$ [S15].
- **SDE:** stochastic differential equation [S5].
- **Vasicek model:** OU process for the short rate (1977) [S15].

5 Self-test

Questions

Q1 ([S5]). Derive the SDEs of the six processes and say which are martingales.

Q2 ([S6]). Under Bachelier, find $\mathbb{P}(S_T < 0)$ for $S_0 = 100$, $\sigma = 30$, $T = 4$.

Q3 ([S7–S8]). Derive the Bachelier call formula and its ATM value. Price $S_0 = K = 100$, $\sigma = 20$, $T = 1$, $r = 0$.

Q4 ([S8]). Give the Bachelier put formula.

Q5 ([S11]). Solve $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$.

Q6 ([S12–S14]). Derive $V_c = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$. What is $\Phi(d_2)$?

Q7. $S_0 = 100$, $K = 105$, $T = 1$, $r = 5\%$, $\sigma = 20\%$: price the call and the put.

Q8 ([S10]). Compare Bachelier and Black–Scholes paths.

Q9 ([S15]). Solve the Vasicek SDE.

Q10 ([S16]). $\kappa = 1.5$, $\theta = 4\%$, $\sigma = 1\%$, $r_0 = 2\%$, $t = 2$: mean, standard deviation and $\mathbb{P}(r_t < 0)$.

Solutions

S1. See the table in §1.2: martingales are (4) $W^2 - t$ and (6) $e^{\theta W - \frac{1}{2}\theta^2 t}$.

S2. $S_T \sim N(100, 3600)$: $\mathbb{P} = \Phi(-100/60) = \Phi(-1.667) \approx 4.8\%$.

S3. See §1.3–1.4. ATM: $20\sqrt{1/2\pi} = 7.979$.

S4. $V_p = e^{-rT}[(K - S_0)\Phi(-d) + \sigma\sqrt{T}\phi(d)]$, $d = \frac{S_0 - K}{\sigma\sqrt{T}}$.

S5. Itô on $\log S$: $\mathrm{d}\log S = (r - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\mathrm{d}W$; $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma W_T}$.

S6. See §1.7–1.8. $\Phi(d_2)$ = risk-neutral probability that the call is exercised.

S7. $d_1 = \frac{\log(100/105) + 0.07}{0.2} = 0.1061$, $d_2 = -0.0939$; $V_c = 100(0.5422) - 105e^{-0.05}(0.4626) = 8.02$. Put: $8.02 - 100 + 105e^{-0.05} = 7.90$.

S8. See §1.5: Black–Scholes moves scale with the price and stay positive; Bachelier moves are level-independent and can go negative.

S9. See §1.9: $r_t = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma\int_0^t e^{\kappa(u-t)}\mathrm{d}W_u$.

S10. Mean $0.02e^{-3} + 0.04(1 - e^{-3}) = 0.0390$; variance $\frac{10^{-4}}{3}(1 - e^{-6}) = 3.325 \times 10^{-5}$, sd 0.577%; $\mathbb{P}(r_t < 0) = \Phi(-0.039/0.00577) = \Phi(-6.76) \approx 0$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Bru & Yor quotation on Doeblin	1.1
S3	Brownian motion and related process classes	1.1
S4	Itô calculus acceptance; McKean	1.1
S5	SDEs and martingales: six examples	1.2
S6	Bachelier model; shortcoming	1.3
S7	Arithmetic BM; call integral; x^*	1.3
S8	Bachelier call formula; ATM	1.4
S9	Black–Scholes model; log return distribution	1.5
S10	Sample paths: Black–Scholes vs Bachelier	1.5
S11	Solving GBM via Itô on $\log S$	1.6
S12	BS call integral; x^*	1.7
S13	Completing the square	1.8
S14	d_1, d_2; BS formula	1.8
S15	Vasicek SDE and solution	1.9
S16	Vasicek mean, variance, distribution	1.10

Appendix B. Flag summary (△!)

1. [S5] Zero drift gives a local martingale; true martingale needs integrability (satisfied by all six examples).
2. [S6–S8] Bachelier with zero drift and e^{-rT} discounting is only risk-neutral if $r = 0$ or S is a forward price.
3. [S10] Same σ used for both models; Bachelier σ is in price units ($\approx \sigma_{\text{BS}} S_0$).
4. [S6] "Reasonable model for interest rates": yes for allowing negative values, but Vasicek-type mean reversion is usually added.
5. [S5, S10] Answers to the examples and the comparison question are not worked on the slides; the working here is my own.